GRIDPRINT analysis report

Verdict

✓

VERIFIED

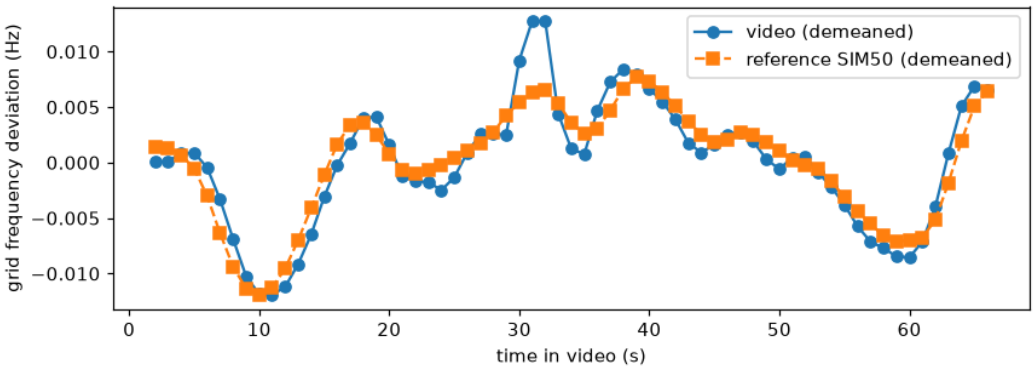
A grid signal is present and matches the reference log at one time. No significant edit evidence was found.

Report state: FINAL

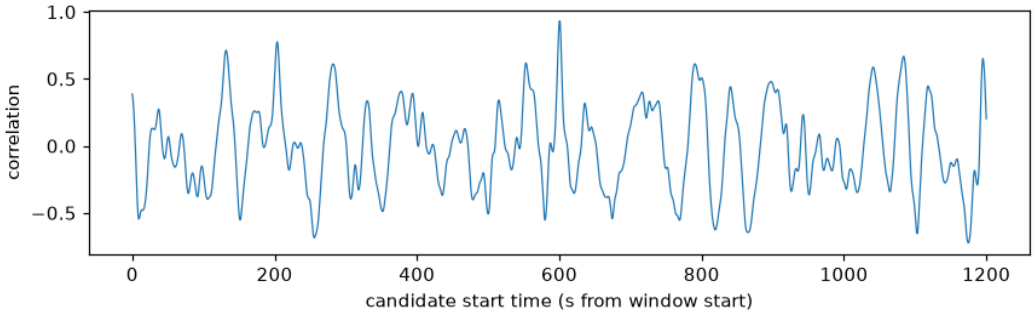
Summary

Estimated recording start (UTC)	2026-01-20T10:00:00.469923+00:00 ± 1.6 s
Grid zone	SIM50
Reasons	• `READOUT_ESTIMATED` — Readout time was estimated from the video (lower confidence). • `MATCH_OK` — The track matches the reference log at one time. • `CLAIM_TIME_MATCH` — The matched time agrees with the claimed time.
Match	peak correlation 0.932, next peak 0.776, track length 68 s

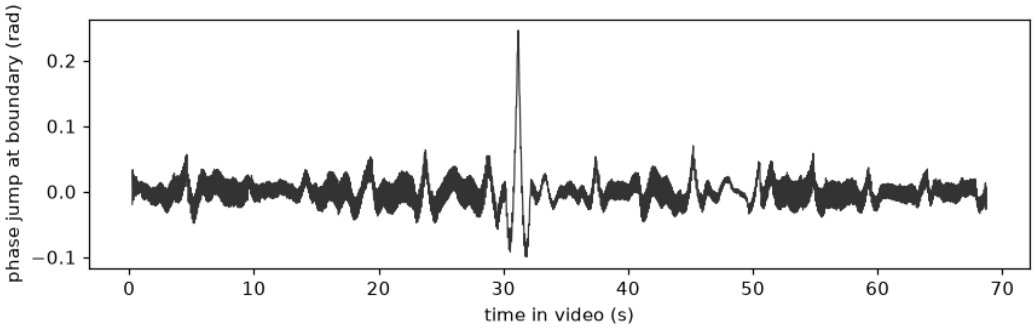
Key plots



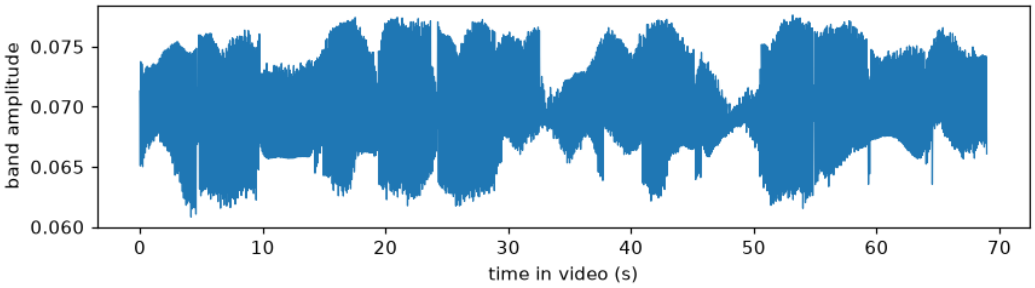
Measured grid frequency variation and matched reference.



Correlation against candidate start times.



Phase jump at frame boundaries. Solid line: edit candidate. Dotted line: timestamp gap.



Flicker band amplitude per frame (fractional modulation).

Limitations

- GRIDPRINT reads the grid frequency from the flicker of mains-powered lights. It is not a deepfake classifier and it is not cryptographic proof.
- A missing signal is never evidence that a video is fake. Many phones hide the flicker ("anti-banding"), and daylight or battery lights do not flicker.
- A time estimate is possible only inside the time range of the reference log for the zone.
- Location output is limited to the grid zone. GRIDPRINT gives no finer location.
- Thresholds come from calibration data. Results outside the calibrated conditions carry more risk of error.
- A CONTRADICTED verdict needs approval by a human reviewer before the report is final.

Provenance

File SHA-256	`d3bae5ec5c6bd5b06af56990a213e435af44e659a5468937053efb4380f4d2db`
Engine version	0.1.0
Config hash	`8f439372b865ee8e`
Thresholds hash	`5c105287d4e1c479`
Job	`aws3-cut-1s-040211`

Appendix

Continuity events

None.

Quality metrics

axis	rows+

band_snr_db	11.374746290115677
bitrate	477539
bits_per_pixel	0.20726519097222224
camera_motion_px	0.033711360290670817
codec	h264
dft_peak_hz	100.03024567608703
estimators_agree	True
foreground_fraction	0.05245036986714976
frames	2070
height	240
jitter_p50_ms	7.105427357601002e-12
jitter_p95_ms	2.042810365310288e-11
motion_level_px	0.11003735380219795
nominal_flicker_hz	100.0
opencv_frame_count	2070
phase_mean_flicker_hz	100.0266658732615
readout_s	0.025285470085468713
readout_source	READOUT_ESTIMATED
rotation_metadata_deg	0
segmentation	ok
static_fraction	1.0
timestamp_source	pts
timing_assumption	pts_is_first_row_exposure_start
track_id	t1
track_samples	65
transfer	bt709
verdict_rule	S13.4
width	320

Stage timings (s)

S1_ingest	0.2943
S2_linear	2.1275
S7_rowsignal	2.5627
S3_readout	4.4826
S4_motion	6.3767
S5_segment	0.9469
S6_patches	3.2888
S8_phase	2.4554
S10_match	0.2329
S11_continuity	0.001